

Sprint 011 Academic Preprint: Uber H3 Index Character Set Verification in Spatial Monads

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Web of Life Simulation Architecture

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Abstract

As the Web of Life simulation architecture evolves to model high-fidelity biogeochemical and spatial domains using Uber H3 discrete global grid systems, strict data validation at spatial boundaries becomes essential. Malformed or invalid H3 index strings can propagate spatial calculation errors across trophic monads and spatial monad stocks. Sprint 011 introduces robust character set verification into `src/spatial/h3_grid.ts` to ensure that all processed Uber H3 index strings strictly conform to valid hexadecimal character sets (`[0-9a-f]`) and expected length constraints. Framed through the lens of systems ecology and thermodynamics, this validation gate acts as an informational filter that preserves mass-energy balance integrity without violating the First and Second Laws of Thermodynamics. Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

1 Introduction & Systems Background

Discrete global grid systems, such as the Uber H3 indexing framework, map continuous biophysical surfaces into hierarchical hexagonal cells. Within the Web of Life simulation engine, trophic energy stocks and biogeochemical mass flows are tightly coupled to these spatial cells.

To maintain system integrity and prevent computational entropy—where malformed spatial pointers corrupt ecological mass balances—telemetry payloads crossing system boundaries must undergo deterministic character set verification. Sprint 011 formalizes this process via the `H3GridManager` class within `src/spatial/h3_grid.ts`.

2 Thermodynamic & Mass-Energy Delta Accounting

In accordance with the governing thermodynamic principles of the Web of Life architecture:

- **Matter Conservation (First Law):** Spatial validation checks operate purely as informational query/filter gates. No matter ($\Delta M = 0 \text{ kg}$), water ($\Delta H_2O = 0 \text{ L}$), or mineral/carbon ($\Delta C, \Delta Min = 0 \text{ mol}$) is created or destroyed during string parsing.
- **Solar Input Only (Second Law):** Computational entropy reduction (sorting and validating spatial indices) is sustained entirely by the deterministic execution cycle driven by external solar and electrical system energy inputs ($\Delta E \approx O(N)$ CPU cycles).

3 Mathematical & Monad Architecture

Let S_{raw} be the incoming spatial telemetry payload containing string I . The Boolean validation function $V(I)$ is defined as:

$$V(I) = \begin{cases} 1 & \text{if } |I| = 15 \text{ and } I \in [0 - 9a - f]^{15} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The state transition for the spatial monad stock register Σ_{spatial} is given by:

$$\Sigma_{\text{spatial}}^{(t+1)} = \begin{cases} \Sigma_{\text{spatial}}^{(t)} \cup \{I\} & \text{if } V(I) = 1 \\ \Sigma_{\text{spatial}}^{(t)} \setminus \{I\} & \text{if } V(I) = 0 \text{ (Rejected / Logged)} \end{cases} \quad (2)$$

3.1 TypeScript Implementation

```
export interface IH3Validator {
  validateIndex(h3Index: string): boolean;
}

export class H3GridManager implements IH3Validator {
  private static readonly H3_REGEX: RegExp = /^[0-9a-f]+$/;
  private static readonly H3_EXPECTED_LENGTH = 15;

  public validateIndex(h3Index: string): boolean {
    if (typeof h3Index !== 'string') return false;
    if (h3Index.length !== H3GridManager.H3_EXPECTED_LENGTH) return
      false;
    return H3GridManager.H3_REGEX.test(h3Index);
  }
}
```

4 Verification and Empirical Testing

Unit tests in `tests/sprint_011.test.ts` verify:

1. Acceptance of valid lowercase 15-character hexadecimal H3 strings.
2. Rejection of uppercase hex strings or incorrect lengths.
3. Rejection of strings containing non-hex characters (`[g-z]`, symbols).
4. Preservation of thermodynamic boundary conditions during rejection logging.

5 Conclusion

Sprint 011 establishes a robust spatial boundary defense mechanism for the Web of Life architecture. By enforcing strict hexadecimal character set verification on Uber H3 indices, the simulation prevents spatial pollution and maintains thermodynamic rigor across trophic monad interactions.

Repository Reference: <https://github.com/pascalranoroarijaona/WebOfLife>